

Phys 224 Problem Set 2

Due Sep. 24 at 11:59 PM. You may talk/consult with whom or whatever, but you must write your own solutions for submission.

1) More algebra practice: A) solve $2(x + 4)(y^2 - 2) + 3y^2 = 9$ for x . Then solve for y .

2) Sin/cos practice: $\sin(x)$ and $\cos(x)$ repeat whenever x increases by 2π radians. How often does $\sin(2\pi\nu t)$ repeat? If t is time, what are the units of ν ?

3) Travelling waves: Let's say we have some function $f(q)$. That is, whenever I feed a value of q into my function, I always get the same $f(q)$ value out. We can have q come from anywhere, including other variables/functions.

A) If we set $f(q) = \sin(q)$, for what values of q will $f(q) = 0$?

B) Now let $q = kx$, so $f(q)$ becomes $\sin(kx)$. For what values of x will we get zero now?

C) Finally, set $q = kx - \omega t$ so our function has become $\sin(kx - \omega t)$. For a given value of the time t , now what values of x give zero?

4) More sin/cos: We'll do this as an in-class exercise on Tuesday, but you should still write down the answer. Let's take two sine waves moving in opposite directions: $f_r = \sin(kx - \omega t)$ is moving right while $f_l = \sin(kx + \omega t)$ is moving left. What is the sum of these two waves $f_l + f_r$? Express your answer as a product of a function of kx and a function of ωt .

5) Take the same f_l and f_r from the previous problem and plug them into desmos for $k = \omega = 1$. Plot f_l , f_r , and $f_l + f_r$ together on the same axes. Show plots of the functions at $t = 0$ and $t = \pi/2$. I suggest you look at the motion for a while to see how two waves moving in opposite directions combine to make something that looks very different. This superposition is a pretty good model for what virtually every instrument does.